Team Name : Stellar Stack

Team Leader Name : Ria Ann Joe

Problem Statement :Optimizing Urban Heat Mitigation and cooling strategies via Artificial Intelligence and Machine Learning (AIML)

Team Members

Team Leader:

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Team Member-1:

Name: Niyati Agarwal
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Name: Hema Akshita
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OPPORTUNITY

Existing Solutions	Proposed Solution
Focus on temperature	Prioritizes vulnerable communities
Generic mitigation	AI-driven zone-specific recommendations
Static reports	Dynamic Digital Twin simulation
One-time prediction	Continuous feedback learning
No optimization	Cost-aware multi-objective optimization

Our platform transforms satellite observations into actionable urban heat mitigation strategies by **identifying high-risk communities, recommending optimized interventions, simulating their impact before implementation**, and **continuously improving recommendations** using real post-implementation satellite observations.

Our solution transforms Earth Observation data into actionable, optimized, and explainable urban heat mitigation plans for city planners.

Unique Selling Proposition

- ❑ **Vulnerability-first prioritization** focusing on people most at risk rather than only the hottest locations.
- ❑ **AI-driven zone-specific recommendations** optimized for cost, impact, and implementation feasibility.
- ❑ **Explainable AI with Confidence and Impact Scores** to ensure transparent, reliable, and evidence-based decisions.
- ❑ **Digital Twin with "What-if" simulation** to evaluate interventions before deployment.
- ❑ **Continuous feedback learning** using updated satellite observations to improve future recommendations.

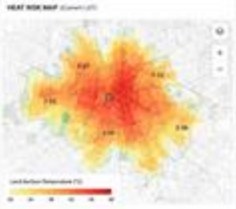
AI Decision Support Framework : Predict → Prioritize → Optimize → Explain → Simulate → Evaluate → Learn

Features

Predict → Prioritize → Optimize → Explain → Simulate → Evaluate → Learn

AI Decision Support Framework → Predict → Prioritize → Optimize → Explain → Simulate → Evaluate → Learn

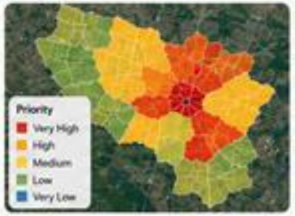
1 Heat Stress Mapping



High-resolution Land Surface Temperature maps from Landsat 8 + ECOSTRESS. Hotspot detection via DBSCAN clustering and UHI intensity index.

Core Feature

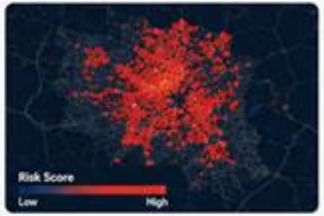
2 Vulnerability Priority Index



Ranks zones by heat exposure combined with social vulnerability — age, income, healthcare access. People-first, not just temperature.

Unique

3 Heat Mortality Risk Score



Estimates lives at risk using temperature, demographics, and healthcare accessibility to quantify human impact.

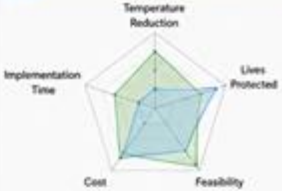
Health-first

4 Zone-specific Recommendations

AI-driven mitigation strategies per zone — green roofs, tree cover, cool pavements — with spatial placement and ΔT estimates.

AI-driven

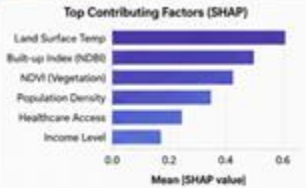
5 Multi-objective Optimization



Balances cost, temperature reduction, lives protected, and feasibility using Pareto-front optimization algorithms.

Optimizer

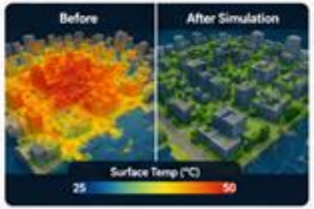
6 Explainable AI



SHAP values explain every recommendation with contributing factors. Confidence scores indicate prediction reliability.

Transparent

7 Digital Twin Simulation



What-if engine simulates interventions before deployment. Continuously improves using post-implementation satellite data.

Simulation

8 Impact Score

-2.8°C	124	18%	High
Temp Reduction	Lives Protected	Energy Saved	Overall Benefit

Estimates expected reduction in temperature (°C), mortality, energy use, and implementation benefits per scenario.

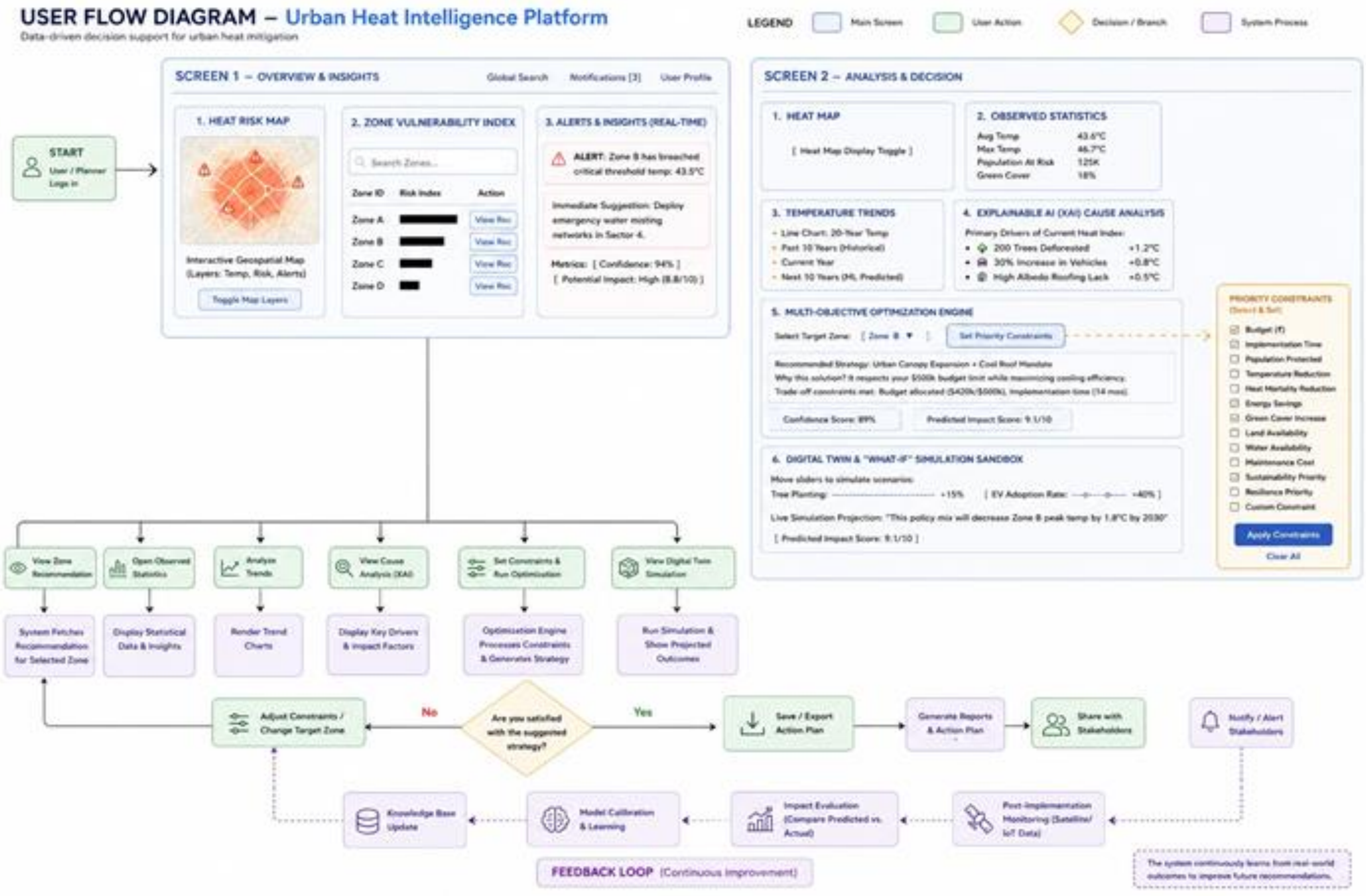
Quantified

9 Early Warning System

Proactive heatwave alerts with recommended emergency actions sent to city officials before dangerous heat events.

Proactive

User Flow Diagram



Process Flow Diagram

1. **Data Collection**
2. **Heat Map Generation**
3. **Heat Prediction**
4. **Historical & Future Heat Trend Analysis**
   (Past 10 Years → Next 10 Years)
5. **Vulnerability Index Calculation**
   (Prioritizes High-Risk Communities)
6. **Heat Mortality Risk Score**
   (Estimates Population at Risk)
7. **Explainable AI**
   (Identifies Key Drivers of Heat Risk)
8. **Multi-objective Optimization**
9. **Zone-specific Mitigation Recommendations**
10. **Recommendation Justification**
    (Explains Selected Strategy & Optimization Trade-offs)
11. **Confidence Score**
    (Prediction & Recommendation Reliability)
12. **Impact Score**
    (Estimated Reduction in Heat Risk & Expected Benefits)
13. **Digital Twin Simulation**
    (Evaluates "What-if" Intervention Scenarios)
14. **Simulation Impact Assessment**
    (Compares Expected Outcomes Before Implementation)
15. **Early Warning & Alert Generation**

Feedback Learning
(Continuously Improves Models Using Real-world Outcomes)

Wireframes

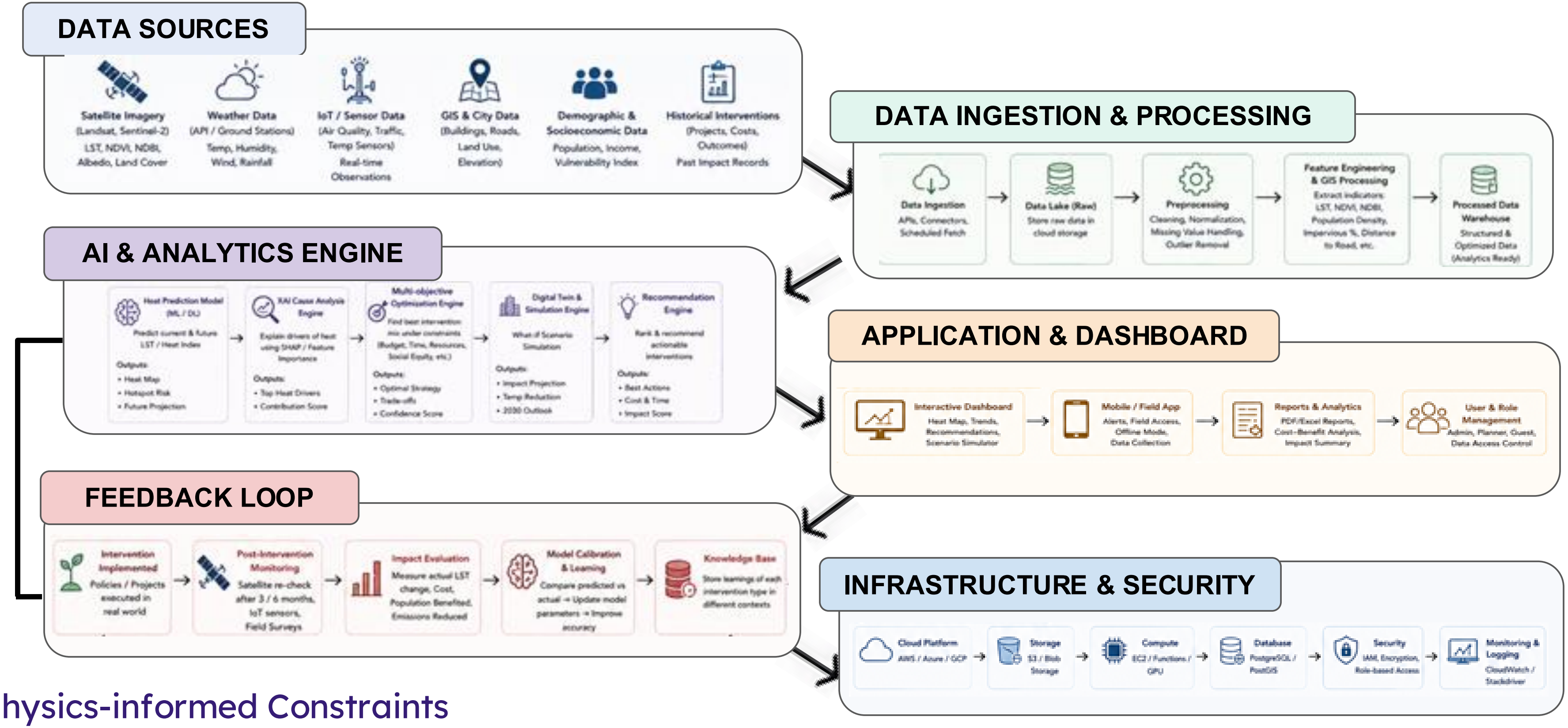
Screen 1

```
+---------------------------------------------------------------------------+
| [Logo]  App Name          [Global Search]      Notifications [3]   User Profile |
+---------------------------------------------------------------------------+
|                                                                           |
|  +-------------------------------+  +----------------------------------+  |
|  | SECTION 1: HEAT RISK MAP      |  | SECTION 2: ZONE VULNERABILITY INDEX |
|  |                               |  |                                  |  |
|  |  +-------------------------+  |  |  [ Search Zones... ]             |  |
|  |  |                         |  |  |                                  |  |
|  |  |      INTERACTIVE        |  |  |  Zone ID   Risk Index    Action  |  |
|  |  |     GEOSPATIAL MAP      |  |  |  ------------------------------  |  |
|  |  | (Layers: Temp, Risk, Alarms) | |  Zone A  [█████████   ] [View Rec] |
|  |  |                         |  |  |  Zone B  [████████████] [View Rec] |
|  |  |                         |  |  |  Zone C  [█████       ] [View Rec] |
|  |  |                         |  |  |  Zone D  [█           ] [View Rec] |
|  |  +-------------------------+  |  +----------------------------------+  |
|  +-------------------------------+                                        |
|                                                                           |
|  +---------------------------------------------------------------------+  |
|  | SECTION 3: CRITICAL ALERTS & INSIGHTS (Real-Time)                   |  |
|  |                                                                     |  |
|  | ⚠ ALERT: Zone B has breached critical threshold! Temp: 43.5°C      |  |
|  | +-----------------------------------------------------------------+ |  |
|  | | Immediate Suggestion: Deploy emergency water misting networks in Sector 4. |
|  | |                                                                 | |  |
|  | | Metrics: [ Confidence Score: 94% ]   [ Potential Impact Score: High (8.8/10) ] |
|  | +-----------------------------------------------------------------+ |  |
|  +---------------------------------------------------------------------+  |
+---------------------------------------------------------------------------+
```

Screen 2

```
+---------------------------------------------------------------------------+
| [Logo]  App Name      < Back to Main Dashboard     Notifications [3]  User Profile |
+---------------------------------------------------------------------------+
|                                                                           |
|  +------------------------------+  +----------------------------------+   |
|  | SECTION 1: HEAT MAP          |  | SECTION 2: OBSERVED STATISTICS   |   |
|  |                              |  |                                  |   |
|  |                              |  | Observed statistics in this particular zone: |
|  |  [ Heat Map Display Toggle ] |  |                                  |   |
|  |                              |  | | Average Current Temperature   [43.6°C] | |
|  |                              |  | | Max Recorded Temperature      [46.7°C] | |
|  |                              |  | | Population At Risk            [125k]   | |
|  |                              |  | | Green Cover                   [18%]    | |
|  +------------------------------+  +----------------------------------+   |
|                                                                           |
|  +------------------------------+  +----------------------------------+   |
|  | SECTION 3:  TEMPERATURE TRENDS |  | SECTION 4: EXPLAINABLE AI (XAI) CAUSE ANALYSIS |
|  |                              |  |                                  |   |
|  | [ Line Chart: 20-Year Temperature ] | Primary Drivers of Current Heat Index: |
|  | - Past 10 Years (Historical Data)   |                                  |   |
|  | - Current Year               |  | | 🌳 200 Trees Deforested nearby    [+1.2°C] | |
|  | - Next 10 Years (ML Predicted Future) | 🚗 30% Increase in Vehicles  [+0.8°C] | |
|  |                              |  | | 🏢 High Albedo Roofing Lack       [+0.5°C] | |
|  +------------------------------+  +----------------------------------+   |
|                                                                           |
|  +--------------------------------------------------------------------+   |
|  | SECTION 5: MULTI-OBJECTIVE OPTIMIZATION ENGINE                     |   |
|  |                                                                    |   |
|  | Select Target Zone: [ Zone B  ▼ ]   Set Priority Constraints: [X] Budget |
|  |                                                                    |   |
|  | Recommended Strategy: Urban Canopy Expansion + Cool Roof Mandate   |   |
|  | - Why this solution? It respects your $500k budget limit while maximizing cooling efficiency. |
|  | - Trade-off constraints met: Budget allocated ($420k/$500k), Implementation time (14 mos). |
|  |                                                                    |   |
|  | [ Confidence Score: 89% ]              [ Predicted Impact Score: 9.1/10 ] |
|  +--------------------------------------------------------------------+   |
|                                                                           |
|  +--------------------------------------------------------------------+   |
|  | SECTION 6: DIGITAL TWIN & "WHAT-IF" SIMULATION SANDBOX             |   |
|  |                                                                    |   |
|  | Move Sliders to Simulate Scenarios:                                |   |
|  | [ Tree Plantings: ----o----------- +15% ]   [ EV Adoption Rate: --------o---- +40% ] |
|  |                                                                    |   |
|  |Live Simulation Projection: "This policy mix will decrease Zone B peak temp by 1.8°C by 2030" |
|  |                                                                    |   |
|  |  [ Predicted Impact Score: 9.1/10 ]                                |   |
|  +--------------------------------------------------------------------+   |
+---------------------------------------------------------------------------+
```

Architecture diagram



Physics-informed Constraints

• NDVI • Surface Albedo • Land Surface Temperature (LST) • Land Use/Land Cover (LULC) • Urban Morphology • Surface Energy Balance

Technologies used:

Layer	Technology	Purpose
Data Sources	Landsat-8/9, Sentinel-2, Weather API, Census, GIS Data	Collect satellite imagery, weather, land use, and demographic data for heat analysis.
Data Processing	Python, Pandas, NumPy, Rasterio, GeoPandas	Clean, preprocess, fuse, and extract geospatial features from satellite and GIS datasets.
AI & ML	XGBoost, LSTM/TFT, U-Net, SHAP	XGBoost – Heat hotspot & risk prediction LSTM/TFT – Heat stress forecasting U-Net – Land Use/Land Cover segmentation SHAP – Explainable AI for model insights
Optimization	NSGA-II (PyMOO)	Generate optimal cooling strategies by balancing temperature reduction, cost, lives protected, and feasibility.
Digital Twin	GeoPandas, Folium	Simulate "what-if" urban cooling scenarios and visualize intervention impacts.
Database	PostgreSQL + PostGIS	Store geospatial datasets, predictions, intervention plans, and historical observations.
Backend	FastAPI	Serve APIs connecting AI models, optimization engine, and dashboard.
Frontend	React, Leaflet	Build an interactive web dashboard with heat maps, recommendations, and simulations.
Deployment	Docker, AWS	Containerize and deploy the application for scalable and reliable cloud hosting.

BHARATIYA ANTARIKSH HACKATHON

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THANK YOU